1. Let $x_1 = -\sqrt{3}$, $x_2 = -\sqrt{3+\sqrt{3}}$, $x_3 = -\sqrt{3+\sqrt{3+\sqrt{3}}}$, ... Evaluate $\lim_{n\to\infty} x_n$:

यदि $x_1 = -\sqrt{3}$, $x_2 = -\sqrt{3+\sqrt{3}}$, $x_3 = -\sqrt{3+\sqrt{3+\sqrt{3}}}$, ... हो, तो $\lim_{n\to\infty} x_n$ का मान ज्ञात कीजिए :

Ans. $\frac{-1-\sqrt{13}}{2}$

Sol. $x_1 = -\sqrt{3}$ $\quad x_2 = -\sqrt{3+\sqrt{3}}$

$x_3 = -\sqrt{3+\sqrt{3+\sqrt{3}}}$

$\lim_{n\to\infty} x_n = x_\infty = x$ $\quad$ Let $= -\sqrt{3+\sqrt{3+\sqrt{3+.....}}}$

$\Rightarrow x = -\sqrt{3-x}$

$\Rightarrow x^2 + x - 3 = 0$ $\quad x = \frac{-1\pm\sqrt{13}}{2}$

Also $x_i < 0$ $\quad \Rightarrow x_\infty = x < 0$

hence $\quad x = \frac{-1-\sqrt{13}}{2}$

2. Let $L = \lim_{x\to0} \frac{e^{x\sin(2007)x} - 1}{x\,\ell n(1+x)}$ and $M = \lim_{x\to\infty} x\left(2x - \left(\sqrt[3]{x^3+x^2+1} + \sqrt[3]{x^3-x^2+1}\right)\right)$ then find the value of LM.

यदि $L = \lim_{x\to0} \frac{e^{x\sin(2007)x} - 1}{x\,\ell n(1+x)}$ और $M = \lim_{x\to\infty} x\left(2x - \left(\sqrt[3]{x^3+x^2+1} + \sqrt[3]{x^3-x^2+1}\right)\right)$ तो LM का मान बताओं।

Ans. 446

Sol. $L = \lim_{x\to0} \frac{e^{x\sin(2007)x} - 1}{x\cdot\ell n(1+x)}$

$L = \lim_{x\to0} \frac{e^{x\cdot\sin(2007)x} - 1}{x\cdot\sin(2007)x} \cdot \frac{\sin(2007)x}{2007x} \frac{(2007)x}{\ell n(1+x)}$

$L = .1.1..2007 = 2007$

$M = \lim_{x\to\infty} x\left[2x - \left(x^3+x^2+1\right)^{\frac{1}{3}} - \left(x^3-x^2+1\right)^{\frac{1}{3}}\right]$

$$= \lim_{x\to\infty} x^2\left[2-\left(1+\frac{1+x^2}{x^3}\right)^{\frac{1}{3}}-\left(1+\frac{1-x^2}{x^3}\right)^{\frac{1}{3}}\right]$$

$$= \lim_{x\to\infty} x^2\left[2-1-\frac{1}{3}a+\frac{2}{9}\frac{1}{2!}a^2-1-\frac{1}{3}b+\frac{2}{9}\frac{1}{2!}b^2\right]$$

$$\begin{cases} a=\dfrac{1+x^2}{x^3} \\ b=\dfrac{1-x^2}{x^3} \end{cases}$$

$$= \lim_{x\to\infty} x^2\left[-\frac{1}{3}\frac{1+x^2}{x^3}-\frac{1}{3}\frac{1-x^2}{x^2}+\frac{2}{9}a^2+\frac{2}{9}b^2\right]$$

$$= \lim_{x\to\infty} -\frac{2}{3}\frac{x^2}{x^3}+\frac{1}{9}\cdot x^2\left\{\frac{\left(1+x^2\right)^2}{x^6}+\frac{\left(1-x^2\right)^2}{x^6}\right\}$$

$$= 0+\frac{2}{9}\frac{1+x^4}{x^4}+0=\frac{2}{9}$$

$$LM = (2007)\ \frac{2}{9} = 446$$

3. Let $L = \lim_{n\to\infty} \dfrac{1}{\frac{1}{\sqrt{2}}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}}\ldots\ldots\ldots \text{n terms}}$. Find the value of $\left(2\sqrt{3}\cot L\right)$.

माना $L = \lim_{n\to\infty} \dfrac{1}{\frac{1}{\sqrt{2}}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}}\ldots\ldots\ldots \text{n पद}}$. $\left(2\sqrt{3}\cot L\right)$ का मान होगा।

Ans. 0

Sol. $T_1 = \dfrac{1}{\sqrt{2}} = \cos\theta \qquad \theta = \dfrac{\pi}{4}$

$$T_2 = \sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}} = \sqrt{\frac{1+\cos\theta}{2}} = \cos\frac{\theta}{2}$$

$$T_3 = \sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}} = \sqrt{\frac{1+\cos\frac{\theta}{2}}{2}} = \cos\left(\frac{\theta}{2^2}\right)$$

$$T_n = \cos\left(\frac{\theta}{2^{n-1}}\right)$$

$$L = \lim_{n\to\infty} \frac{1}{T_1, T_2, ...T_n}$$

$$L = \lim_{n\to\infty} \frac{1}{\cos\theta \cdot \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2^5}\right).....\cos\left(\frac{\theta}{2^{n-1}}\right)}$$

$$\lim_{n\to\infty} \frac{1}{\frac{\sin\left(2^n \cdot \frac{\theta}{2^{n-1}}\right)}{2^n \cdot \sin\left(\frac{\theta}{2^{n-1}}\right)}}$$

$$= \lim_{n\to\infty} \frac{2^n \cdot \sin\left(\frac{\theta}{2^{n-1}}\right)}{\sin 2\theta}$$

$$= \lim_{n\to\infty} \frac{2\theta}{\sin 2\theta} \cdot \frac{\sin\left(\frac{\theta}{2^{n-1}}\right)}{\left(\frac{\theta}{2^{n-1}}\right)}$$

$$L = \frac{2\theta}{\sin 2\theta}; \qquad \theta = \frac{\pi}{4}$$

$$L = \frac{\pi}{2} \qquad \cot L = 0$$

$$2\sqrt{3}\cot L = 0$$

4. $\lim\limits_{x\to\infty} x^2 \sin\left(\ell n \sqrt{\cos\frac{\pi}{x}}\right).$

Ans. $-\frac{\pi^2}{4}$

Sol. $t = \lim\limits_{x\to\infty} x^2 \sin\left(\ell n\sqrt{\cos\left(\frac{\pi}{x}\right)}\right)$

put $x = \frac{1}{y}$ as $x \to \infty,$ $y \to O^+$

$$\Rightarrow t=\lim_{y\to 0}\frac{\sin\left\{\frac{1}{2}\ell n\left(\cos \pi y\right)\right\}}{\frac{1}{2}\ell n\left(\cos \pi y\right)}\cdot\frac{1}{2}\frac{\ell n\cos \pi y}{y^2}$$

$$\Rightarrow t=\lim_{y\to 0}\frac{1}{2}\cdot\frac{\ell n\left[1-\{1-\cos \pi y\}\right]}{1-\cos \pi y}\cdot\frac{1-\cos \pi y}{\left(\pi y\right)^2}\pi^2$$

$$\Rightarrow t=\frac{1}{2}(-1)\frac{1}{2}\cdot\pi^2$$

$$\Rightarrow t=\frac{-\pi^2}{4}$$

5. $\lim\limits_{x\to\infty}\left[\cos\left(2\pi\left(\frac{x}{1+x}\right)^a\right)\right]^{x^2}$ $a\in Q$

Ans. $e^{-2\pi^2a^2}$

Sol. $L=\lim\limits_{x\to\infty}\left[\cos\left\{2\pi\left(\frac{x}{1+x}\right)^a\right\}\right]^{x^2}$ $a\in Q$

$$L=e^{\lim\limits_{x\to\infty}\left[\cos 2\pi\left(\frac{x}{1+x}\right)^a-1\right]\cdot x^2}$$

$$L=e^{\lim\limits_{x\to\infty}\frac{\cos\left\{2\pi-2\pi\left(\frac{\pi}{1+x}\right)^a\right\}-1}{4\pi^2\left[1-\left(\frac{x}{1+x}\right)^a\right]^2}\cdot 4\pi^2\cdot x^2\left[1-\left(\frac{x}{1+x}\right)^a\right]^2}$$

Put $x=\frac{1}{y}$, $x\to\infty$, $y\to 0^+$

$$L=e^{-\frac{4}{2}\pi^2\lim\limits_{y\to 0}\frac{1}{y^2}\left[1-(1+y)^{-a}\right]^2}$$

$$=e^{-2\pi^2\lim\limits_{y\to 0}\frac{\left[1-ay-\frac{(-a)(-a-1)}{21}y^2.....\right]^2}{y^2}}$$

$$=e^{-2\pi^2a^2}$$

6. Let $f(x)=\frac{\sin^{-1}(1-\{x\}).\cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}\,.\,(1-\{x\})}$ then find $\lim\limits_{x\to 0^+}$ f(x) and $\lim\limits_{x\to 0^-}$ f(x), where $\{x\}$ denotes the fractional part function.

माना $f(x)=\frac{\sin^{-1}(1-\{x\}).\cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}\,.\,(1-\{x\})}$ तो $\underset{x\to0^+}{\text{Lim}}\ f(x)$ तथा $\underset{x\to0^-}{\text{Lim}}\ f(x)$ का मान ज्ञात करो जहाँ पर $\{x\}$ भिन्नांश फलन है।

Ans. $\frac{\pi}{2}, \frac{\pi}{2\sqrt{2}}$

Sol. $\text{R.H.L.}=\lim\limits_{x\to0^+}\frac{\sin^{-1}(1-x)\cos^{-1}(1-x)}{\sqrt{2x}(1-x)}$

$=\lim\limits_{x\to0^+}\frac{\sin^{-1}(1-x)}{(1-x)}\frac{\sin^{-1}\left[\sqrt{1-(1-x)^2}\right]}{\sqrt{2x-x^2}}\frac{\sqrt{2x-x^2}}{\sqrt{2x}}$

$=\frac{\pi}{2}\cdot(1)\cdot(1)$

$=\frac{\pi}{2}$

$\text{L.H.S.}=\lim\limits_{x\to0^-}\frac{\sin^{-1}(1-(x+1))}{(1-(x+1))}\frac{\cos^{-1}(1-(1+x))}{\sqrt{2(1+x)}}$

$\lim\limits_{x\to0^-}\frac{\sin^{-1}(-x)}{(-x)}\cdot\frac{\cos^{-1}(-x)}{\sqrt{2(1+x)}}$

$-x=h,\ x\to0^-,\ h\to0^+$

$=\lim\limits_{h\to0}\frac{\cos^{-1}(h)}{\sqrt{2}\sqrt{1-h}}$

$=\frac{1}{\sqrt{2}}\lim\limits_{h\to0}\frac{\cos^{-1}(h)}{\sqrt{1-h}}=\frac{1}{\sqrt{2}}\frac{\pi}{2}$

7. $\underset{n\to\infty}{\text{Lim}}\left(\frac{\sqrt{n^2+n}-1}{n}\right)^{2\sqrt{n^2+n}-1}$

Ans. e^{-1}

Sol. $\lim\limits_{n\to\infty}\left(\frac{\sqrt{n^2+n}-1}{n}\right)^{2\sqrt{n^2+n}-1}$ (1^∞)

$=e^{\lim\limits_{n\to\infty}\left[\frac{\sqrt{n^2+n}-(n+1)}{n}\right]\left[2\sqrt{n^2+n}-1\right]}$

$$= e^{\lim_{n\to\infty} \frac{-n-1}{n\left[\sqrt{n^2+n}+n+1\right]} \frac{4\left(n^2+n\right)-1}{2\sqrt{n^2+n}+1}}$$

$$= e^{\lim_{n\to\infty} \frac{-(n+1)}{n} \frac{4\left(n^2+n\right)-1}{\left[\sqrt{n^2+n}+n+1\right]\left[2\sqrt{n^2+n}+1\right]}}$$

$$= e^{-1}$$

8. $\lim\limits_{x\to a} \dfrac{1}{(a^2-x^2)^2}\left(\dfrac{a^2+x^2}{ax} - 2\sin\left(\dfrac{a\pi}{2}\right)\sin\left(\dfrac{\pi x}{2}\right)\right)$ where a is an odd integer.

$\lim\limits_{x\to a} \dfrac{1}{(a^2-x^2)^2}\left(\dfrac{a^2+x^2}{ax} - 2\sin\left(\dfrac{a\pi}{2}\right)\sin\left(\dfrac{\pi x}{2}\right)\right)$ जहाँ a एक विषम पूर्णांक है।

Ans. $\dfrac{\pi^2a^2+4}{16a^4}$

Sol. $\lim\limits_{x\to a} \dfrac{1}{(a+x)^2 ax} \qquad \dfrac{a^2+x^2-2ax.\sin\left(\dfrac{a\pi}{2}\right)\sin\left(\dfrac{\pi x}{2}\right)}{(a-x)^2}$

$$= \frac{1}{4a^4}\lim_{x\to a}\frac{a^2+x^2-2a\cdot\sin\left(\dfrac{a\pi}{2}\right)x\cdot\sin\dfrac{\pi}{2}x}{(a-x)^2}$$

By L – Hospital rule

$$= \frac{1}{4a^4}\lim_{x\to a}\frac{x-a\sin\left(\dfrac{a\pi}{2}\right)\left\{\sin\dfrac{\pi}{2}x+\dfrac{\pi}{2}x\cdot\cos\dfrac{\pi}{2}x^4\right\}}{x-a}$$

Again L – Hospital rule

$$= \frac{1}{4a^4}\lim_{x\to a}\left[1-a\sin\left(\frac{a\pi}{2}\right)\left\{\frac{\pi}{2}\cos\frac{\pi}{2}x+\frac{\pi}{2}\cos\frac{\pi}{2}x-\frac{\pi^2}{4}x\sin\frac{\pi}{2}x\right\}\right]$$

$$= \frac{1}{4a^4}\left[1-a\sin\frac{a\pi}{2}\left(\pi.\cos\frac{\pi}{2}a-\frac{\pi^2}{4}a\sin\frac{\pi}{2}a\right)\right]$$

$$= \frac{1}{4a^4}\left[1+\frac{a^2\pi^2}{4}\sin^2\left(\frac{\pi}{2}a\right)\right]$$

$$= \frac{4+(a\pi)^2}{16a^4}$$

9. If $L = \lim\limits_{x\to 1} \dfrac{(1-x)(1-x^2)(1-x^3).....(1-x^{2n})}{[(1-x)(1-x^2)(1-x^3).....(1-x^n)]^2}$ then show that L can be equal to :

(a) $\prod\limits_{r=1}^{n}\dfrac{n+r}{r}$ (b) $\dfrac{1}{n!}\prod\limits_{r=1}^{n}(4r-2)$

(c) The sum of the coefficients of two middle terms in the expansion of $(1+x)^{2n-1}$.

(d) The coefficient of x^n in the expansion of $(1+x)^{2n}$.

यदि $L = \lim\limits_{x\to 1} \dfrac{(1-x)(1-x^2)(1-x^3).....(1-x^{2n})}{[(1-x)(1-x^2)(1-x^3).....(1-x^n)]^2}$ तो सिद्ध करो कि L निम्नलिखित के बराबर हो सकता है :

(a) $\prod\limits_{r=1}^{n}\dfrac{n+r}{r}$ (b) $\dfrac{1}{n!}\prod\limits_{r=1}^{n}(4r-2)$

(c) $(1+x)^{2n-1}$ के विस्तार में दो मध्य पदों के गुणांकों का योग

(d) $(1+x)^{2n}$ के विस्तार में x^n का गुणांक

Ans.

Sol. $L = \lim\limits_{x\to 1}\dfrac{\left(1-x^{n+1}\right)\left(1-x^{n+2}\right).....\left(1-x^{2n}\right)}{(1-x)\left(1-x^2\right).....\left(1-x^n\right)}$

$L = \lim\limits_{x\to 1}\dfrac{1-x^{n+1}}{1-x}\cdot\dfrac{1-x^{n+2}}{1-x^2}\cdot\dfrac{1-x^{2n}}{1-x^n}$

$= \dfrac{(n+1)}{1}\dfrac{(n+2)}{2}.....\dfrac{(2n)}{n} = \prod\limits_{r=1}^{n}\dfrac{n+r}{r}$ (a)

$= \dfrac{1\cdot 2\cdot.....\cdot n}{1\cdot 2\cdot 3\cdot.....n}\cdot\dfrac{(n+1).....(2n)}{1\cdot 2\cdot.....n} = \dfrac{(2n)!}{(n!)^2} = {}^{2n}C_n$ _____(1)

(c) $(1+x)^{2n-1} \rightarrow$ cofficients of middle term $= {}^{2n-1}C_{n-1}$ and ${}^{2n-1}C_n$ (c)

sum $= {}^{2n-1}C_{n-1} + {}^{2n-1}C_n = {}^{2n}C_n$

from (1)

$\Rightarrow \dfrac{\left[1\cdot 3\cdot 5\cdot.....(2n-1)\right]}{n!n!}(2\cdot 4\cdot.....2n)$

$\Rightarrow \dfrac{2\cdot 6\cdot 10.....(4n-2)}{1\cdot 2\cdot 3\cdot\quad n}\cdot\dfrac{(1\cdot 2\cdot....n)}{n!}$

$\prod\limits_{r=1}^{n}\dfrac{4r-2}{n!} = \dfrac{1}{n!}\prod\limits_{r=1}^{n}(4r-2)$ (b)

10. Let $x_0 = 2\cos\frac{\pi}{6}$ and $x_n = \sqrt{2+x_{n-1}}$, $n = 1, 2, 3, ...$, find $\lim\limits_{n\to\infty} 2^{(n+1)}\cdot\sqrt{2-x_n}$.

माना कि $x_0 = 2\cos\frac{\pi}{6}$ तथा $x_n = \sqrt{2+x_{n-1}}$, $n = 1, 2, 3, ...$, तो $\lim\limits_{n\to\infty} 2^{(n+1)}\cdot\sqrt{2-x_n}$ का मान बताओं।

Ans. $\frac{\pi}{3}$

Sol. $x_0 = 2\cos\frac{\pi}{6}$

$$x_1 = \sqrt{2+2\cos\frac{\pi}{6}} = 2\cos\left(\frac{\pi}{6}\cdot\frac{1}{2}\right)$$

$$x_2 = \sqrt{2+2\cos\left(\frac{\pi}{6}\cdot\frac{1}{2}\right)} = 2\cos\left(\frac{\pi}{6}\cdot\frac{1}{2^2}\right)$$

llly $x_n = 2\cos\left(\frac{\pi}{6}\frac{1}{2^n}\right)$

Now $\lim\limits_{n\to\infty} 2^{n+1}\sqrt{2-2\cos\frac{\theta}{2^n}}$ $\qquad\qquad \theta = \frac{\pi}{6}$

$$= \lim_{n\to\infty} 2^{n+1}2\cdot\sin\left(\frac{\theta}{2^{n+1}}\right)$$

$$= \lim_{n\to\infty} 2\theta\cdot\frac{\sin\left(\frac{\theta}{2^{n+1}}\right)}{\left(\frac{\theta}{2^{n+1}}\right)} = 2\theta = 2\left(\frac{\pi}{6}\right) = \frac{\pi}{3}$$

11. $\lim\limits_{x\to0}\left[\frac{\ell n\,(1+x)^{1+x}}{x^2} - \frac{1}{x}\right]$.

Ans. 1/2

Sol. $\lim\limits_{x\to0}\frac{(1+x)\ell n(1+x)-x}{x^2}$

$$= \lim_{x\to0}\frac{(1+x)\left[x-\frac{x^2}{2}+\frac{x^3}{3}.....\right]-x}{x^2} = \frac{1}{2}$$

12. Let $L=\prod_{n=3}^{\infty}\left(1-\frac{4}{n^2}\right)$; $M=\prod_{n=2}^{\infty}\left(\frac{n^3-1}{n^3+1}\right)$ and $N=\prod_{n=1}^{\infty}\frac{(1+n^{-1})^2}{1+2n^{-1}}$, then find the value of $L^{-1}+M^{-1}+N^{-1}$.

माना कि $L=\prod_{n=3}^{\infty}\left(1-\frac{4}{n^2}\right)$; $M=\prod_{n=2}^{\infty}\left(\frac{n^3-1}{n^3+1}\right)$ तथा $N=\prod_{n=1}^{\infty}\frac{(1+n^{-1})^2}{1+2n^{-1}}$ तो $L^{-1}+M^{-1}+N^{-1}$ का मान ज्ञात कीजिए।

Ans. 8

Sol. $L=\prod_{n=3}^{\infty}\frac{(n-2)(n+2)}{n\quad n}=\lim_{k\to\infty}\prod_{n=3}^{k}\frac{n-2}{n}\cdot\frac{n+2}{n}$

$$L=\lim_{k\to\infty}\left(\frac{1}{3}\cdot\frac{5}{3}\right)\left(\frac{2}{4}\cdot\frac{6}{4}\right)\left(\frac{3}{5}\cdot\frac{7}{5}\right)\cdot\left(\frac{4}{6}\cdot\frac{8}{6}\right).....\left(\frac{k-2}{k^3}\cdot\frac{k+2}{k}\right)$$

$$=\lim_{k\to\infty}\left(\frac{1}{3}\right)\left(\frac{2}{4}\right).....\left(\frac{k-3}{k-1}\right)\left(\frac{k+2}{k}\right)$$

$$L=\frac{1}{6}$$

$$M=\lim_{k\to\infty}\prod_{n=2}^{k}\frac{(n-1)}{(n+1)}\frac{\left(n^2+n+1\right)}{\left(n^2-n+1\right)}$$

$$=\lim_{k\to\infty}\left(\frac{1}{3}\cdot\frac{7}{3}\right)\left(\frac{2}{4}\cdot\frac{13}{7}\right)\cdot\left(\frac{3}{5}\cdot\frac{21}{13}\right).....\left(\frac{k-1}{k+1}\cdot\frac{k^2+k+1}{k^2-k+1}\right)$$

$$=\lim_{k\to\infty}\frac{1\cdot2}{3}\cdot\frac{k^2+k+1}{k\cdot(k+1)}=\frac{2}{3}$$

$$N=\lim_{k\to\infty}\prod_{n=1}^{k}\frac{(n+1)^2}{n(n+2)}=\lim_{k\to\infty}\left(\frac{2}{1}\cdot\frac{2}{3}\right)\left(\frac{3}{2}\cdot\frac{3}{4}\right)\cdot\left(\frac{4}{3}\cdot\frac{4}{5}\right).....\left(\frac{k}{k-1}\cdot\frac{k}{k+1}\right)\left(\frac{k+1}{k}\cdot\frac{k+1}{k+2}\right)$$

$$\lim_{k\to\infty}\frac{2}{1}\cdot\frac{k+1}{k+2}=2$$

$$L^{-1}+M^{-1}+N^{-1}=6+\frac{3}{2}+\frac{1}{2}=8$$

13. Let $f(x)=\lim_{n\to\infty}\sum_{n=1}^{n}3^{n-1}\sin^3\frac{x}{3^n}$ and $g(x)=x-4f(x)$. Evaluate $\lim_{x\to0}(1+g(x))^{\cot x}$.

माना कि $f(x)=\lim_{n\to\infty}\sum_{n=1}^{n}3^{n-1}\sin^3\frac{x}{3^n}$ तथा $g(x)=x-4f(x)$ तो $\lim_{x\to0}(1+g(x))^{\cot x}$ का मान ज्ञात कीजिए।

Ans. $g(x) = \sin x$ and $l = e$

Sol. $$f(x) = \lim_{n\to\infty} 3^{n-1} \frac{3\sin\left(\frac{x}{3^n}\right) - \sin\left(\frac{x}{3^{n-1}}\right)}{4}$$

$$= \frac{1}{4}\lim_{n\to\infty}\left[\left(3\cdot\sin\left(\frac{x}{3}\right) - \sin x\right) + \left(3^2\cdot\sin\left(\frac{x}{3^2}\right) - 3\sin\left(\frac{x}{3}\right)\right) + ... + \left(3^n \sin\left(\frac{x}{3^n}\right) - 3^{n-1}\sin\left(\frac{x}{3^{n-1}}\right)\right)\right]$$

$$= \frac{1}{4}\lim_{n\to\infty}\left[3^n \sin\left(\frac{x}{3^n}\right) - \sin x\right]$$

$$= \frac{1}{4}\left[\lim_{n\to\infty} x\cdot\frac{\sin\left(\frac{x}{3^n}\right)}{\left(\frac{x}{3^n}\right)} - \sin x\right]$$

$$f(x) = \frac{1}{4}(x - \sin x)$$

$$4f(x) = x - \sin x \qquad x - 4f(x) = \sin x = g(x)$$

Now $\lim_{x\to 0}\left[1 + g(x)\right]^{\cot x} = e^{\lim_{x\to 0} \sin x\cdot\cot x} = e$

$\Rightarrow \ell = e$

14. If $f(n, \theta) = \prod_{r=1}^{n}\left(1 - \tan^2\frac{\theta}{2^r}\right)$, then compute $\lim_{n\to\infty} f(n,\theta)$

यदि $f(n, \theta) = \prod_{r=1}^{n}\left(1 - \tan^2\frac{\theta}{2^r}\right)$, तो $\lim_{n\to\infty} f(n,\theta)$ का मान ज्ञात कीजिए।

Ans. $\frac{\theta}{\tan\theta}$

Sol. $1 - \tan^2 x = (1 + \tan^2 x)\cos 2x = \frac{\cos 2x}{\cos^2 x}$

Now $f(n,\theta) = \prod_{r=1}^{n}\left(1 - \tan^2\frac{\theta}{2^r}\right)$

$$= \prod_{r=1}^{n}\frac{\cos\left(\frac{\theta}{2^{r-1}}\right)}{\cos^2\left(\frac{\theta}{2^r}\right)}$$

$$= \left(\frac{\cos\theta}{\cos\frac{\theta}{2}\cdot\cos\frac{\theta}{2}}\right)\left(\frac{\cos\frac{\theta}{2}}{\cos\left(\frac{\theta}{2^2}\right)\cdot\cos\left(\frac{\theta}{2^2}\right)}\right)\left(\frac{\cos\frac{\theta}{2^2}}{\cos\frac{\theta}{2^3}\cdot\cos\frac{\theta}{2^3}}\right).....\left(\frac{\cos\frac{\theta}{2^{n-1}}}{\cos\frac{\theta}{2^n}\cos\frac{\theta}{2^n}}\right)$$

$$= \frac{\cos\theta}{\cos\left(\frac{\theta}{2}\right)\cdot\cos\left(\frac{\theta}{2^2}\right)\cdot\cos\left(\frac{\theta}{2^n}\right)} = \frac{\cos\theta}{\frac{\sin\left(2^n\cdot\frac{\theta}{2^n}\right)}{2^n\sin\left(\frac{\theta}{2^n}\right)}}$$

$$f(n,\theta) = \frac{\cos\theta}{\sin\theta}\theta.\frac{\sin\left(\frac{\theta}{2^n}\right)}{\left(\frac{\theta}{2^n}\right)}$$

$$\lim_{n\to\infty} f(n,\theta) = \frac{\theta}{\tan\theta}$$

15. If $\lim\limits_{x\to0^+} \ell n(x\,\ell n\,a)\ \ell n\left(\frac{\ell n\,ax}{\ell n\frac{x}{a}}\right) = 6,$ then find the value of a.

यदि $\lim\limits_{x\to0^+} \ell n(x\,\ell n\,a)\ \ell n\left(\frac{\ell n\,ax}{\ell n\frac{x}{a}}\right) = 6$ तो a का मान ज्ञात कीजिये।

Ans. e^3

Sol. $x \to 0^+$ $\quad t = \ell n\,x \to \infty$ $\quad \ell n(\ell n\,a) = b$ $\quad \ell n\,a = A$

Now $\lim\limits_{x\to0^+}\left[\ell n\,x + \ell n(\ell n\,a)\right]\ell n\left[\frac{\ell n\,a + \ell n\,x}{\ell n\,x - \ell n\,a}\right] = 6$

$$= \lim_{t\to\infty}\left[t + \ell n\,A\right]\ell n\left[\frac{A+t}{t-A}\right] = 6$$

$$\Rightarrow \lim_{t\to\infty}\frac{t+\ell n\,A}{t}\cdot A\,\frac{\ell n\left(1+\frac{A}{t}\right) - \ell n\left(1-\frac{A}{t}\right)}{\frac{A}{t}} = 6$$

$\Rightarrow 1.A\left[1-(-1)\right]=6$

$\Rightarrow A=3 \qquad \Rightarrow \ell n\, a=3 \qquad \Rightarrow a=e^3$

16. Evaluate $\lim\limits_{x\to\infty}\left(\frac{x}{e}-x\left(\frac{x}{x+1}\right)^x\right)$.

$\lim\limits_{x\to\infty}\left(\frac{x}{e}-x\left(\frac{x}{x+1}\right)^x\right)$ का मान ज्ञात कीजिये।

Ans. $\frac{-1}{2e}$

Sol. $\lim\limits_{x\to\infty}\left[\frac{x}{e}-x\left(\frac{x}{1+x}\right)^x\right]$

$=\lim\limits_{y\to0^+}\left[\frac{1}{ey}-\frac{1}{y}\left(\frac{1}{1+y}\right)^{\frac{1}{y}}\right]$

$=\lim\limits_{y\to0^+}\left[\frac{1}{ey}-\frac{1}{y}e^{-\frac{1}{y}\ell n(1+y)}\right] \qquad a^b=e^{b\ell na}$

$=\lim\limits_{y\to0^+}\left[\frac{1}{ey}-\frac{1}{y}e^{-\frac{1}{y}\left\{y-\frac{y^2}{2}+\frac{y^3}{3}.....\right\}}\right]$

$=\lim\limits_{y\to0^+}\left[\frac{1}{ey}-\frac{1}{y}e^{-1}e^{\left(\frac{y}{2}-\frac{y^2}{3}+.....\right)}\right]$

neglate y^2 and higher power

$=\lim\limits_{y\to0^+}\left[\frac{1}{ey}-\frac{1}{ey}e^{\frac{y}{2}}\right]$

$=\lim\limits_{y\to0^+}\left[\frac{1}{ey}-\frac{1}{ey}\left\{1+\frac{y}{2}+\frac{y^2}{42!}.....\right\}\right]$

$=-\frac{1}{2e}$

17. Through a point A on a circle, a chord AP is drawn & on the tangent at A a point T is taken such that AT = AP. If TP produced meet the diameter through A at Q, prove that the limiting value of AQ when P moves upto A is double the diameter of the circle.

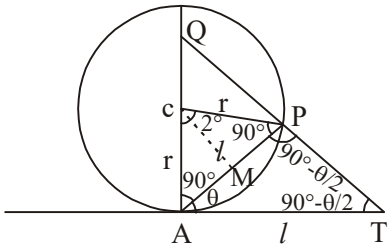
वृत्त पर स्थित एक बिन्दु A से एक जीवा AP खींची गई है तथा A पर स्पर्श रेखा पर एक बिन्दु T इस प्रकार से लिया गया है कि AT = AP. यदि TP रेखा को बढ़ाने पर यह A से खीचीं जाने वाली व्यास को यदि Q पर मिलती हो तो सिद्ध करों कि जब $P \to A$ तो AQ का मान वृत्त की व्यास का दुगना होगा।

Ans.

Sol. ΔACM

$$\sin\theta = \frac{\ell}{2r}$$

$$AP = AT = 2r\sin\theta$$



$\Delta AT\,Q$

$$\tan\left(90° - \frac{\theta}{2}\right) = \frac{AQ}{\ell}$$

$$\Rightarrow AQ = \ell\cot\frac{\theta}{2} = \ell 2r\sin\theta\cot\frac{\theta}{2}$$

$$AQ = 4r\cos^2\frac{\theta}{2}$$

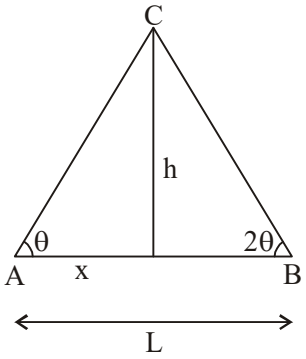
as $P \to A$ $\quad Q \to 0$ $\quad AQ = 4r = 2(2r) = 2$ (diameter)

18. At the end points A, B of the fixed segment of length L, lines are drawn meeting in C and making angles θ and 2θ respectively with the given segment. Let D be the foot of the altitude CD and let x represents the length of AD. Find the value of x as θ tends to zero i.e. $\lim\limits_{\theta\to 0} x$.

L लम्बाई वाले रेखा खण्ड के सिरे A व B से, रेखाखण्ड से क्रमशः θ तथा 2θ कोण बनाती हुयी, रेखा खीची जाती है जो कि C पर मिलती gSA; fn j s[kk[k Mi j C से डाले गये लम्ब CD हो तथा AD की लम्बाई x है तो $\lim\limits_{\theta\to 0} x$ का मान ज्ञात करो।

Ans. $\frac{2L}{3}$

Sol.



$$\frac{hL}{2} = \frac{1}{2} AC.BC.\sin 3\theta$$

$$hL = \frac{1}{2} x \sec\theta \cdot h \cos ec 2\theta \sin 3\theta$$

$$x = \frac{L\cos\theta.\sin 2\theta}{\sin 3\theta}$$

$$\lim_{\theta\to 0} x = L \lim_{\theta\to 0} \frac{3\theta}{\sin 3\theta} \frac{\sin 2\theta}{2\theta} \cdot \frac{2}{3}$$

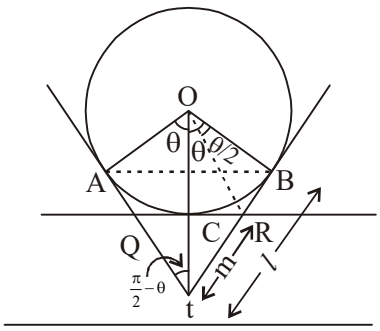
$$= \frac{2L}{3}$$

19. At the end-points and the midpoint of a circular arc AB tangent lines are drawn, and the points A and B are joined with a chord. Prove that the ratio of the areas of the two triangles thus formed tends to 4 as the arc AB decreases indefinitely.

किसी वृत्तीय चाप AB के अन्त बिन्दुओं (End points) तथा मध्य बिन्दु (mid point) पर स्पर्श रेखायें खीची जाती है तथा बिन्दु A तथा Bको एक जीवा के द्वारा मिलाया जाता है। सिद्ध करों कि जब चाप AB की लम्बाई घटते–घटते शून्य पर आयेगी तो इस प्रकार से बने दोनों त्रिभुजो के क्षेत्रफल का अनुपात का मान 4 हो जायेगा।

Ans.

Sol.



RB = l – m

$$\Delta PAB = \frac{\ell^2}{2}\sin 2\theta$$

$$\Delta PQR = \frac{m^2}{2}\sin 2\theta$$

Now

ΔPOB

$$\rightarrow \frac{r}{\ell} = \tan\left(\frac{\pi}{2} - \theta\right)$$

$$\frac{\Delta PAB}{\Delta PQR} = \left(\frac{\ell}{m}\right)^2 \qquad \text{_____(1)}$$

$r = l \cot\theta$ _____(2)

ΔROB

$$\Rightarrow \frac{r}{\ell - m} = \cot\frac{\theta}{2} \qquad \text{_____(3)}$$

$$\Rightarrow r = (\ell - m)\cot\frac{\theta}{2}$$

from (2) and (3)

$$\ell\cot\theta = (\ell - m)\cot\frac{\theta}{2}$$

$$\frac{\cot\theta}{\cot\frac{\theta}{2}} = 1 - \frac{m}{\ell} \qquad \Rightarrow \frac{m}{\ell} = \frac{\cot\frac{\theta}{2} - \cot\theta}{\cot\frac{\theta}{2}}$$

$$= \frac{1}{2\cos^2\frac{\theta}{2}} \qquad \text{_____(4)}$$

from (1) and (4)

$$\frac{\Delta PAB}{\Delta PQR} = 4\cos^4\frac{\theta}{2} \qquad AB \rightarrow \theta \rightarrow O$$

$$\lim_{\theta\to 0}\frac{\Delta PAB}{\Delta PQR} = 4$$

20. If $L = \lim_{x\to 0}\left(\frac{1}{\ell n(1+x)} - \frac{1}{\ell n(x+\sqrt{1+x^2})}\right)$ then find the value of $\frac{L+153}{L}$.

यदि $L = \lim_{x\to 0}\left(\frac{1}{\ell n(1+x)} - \frac{1}{\ell n(x+\sqrt{1+x^2})}\right)$ तो $\frac{L+153}{L}$ का मान ज्ञात कीजिए।

Ans. 307

Sol. $$L = \lim_{x\to 0}\frac{\ell n\left(x+\sqrt{1+x^2}\right) - \ell n(1+x)}{\ell n\left(x+\sqrt{1+x^2}\right)\cdot\frac{\ell n(1+x)}{x}\cdot x}$$

$$= \lim_{x\to 0}\frac{\ell n\left(\sqrt{1+x^2}\right) + \ell n\left(1+\frac{x}{\sqrt{1+x^2}}\right) - \ell n(1+x)}{\left[\ell n\left(\sqrt{1+x^2}\right) + \ell n\left(1+\frac{x}{\sqrt{1+x^2}}\right)\right]x}$$

$$\lim_{x\to 0}\frac{\frac{1}{2}\left\{x^2 - \frac{x^4}{2} + ...\right\}\left\{\frac{x}{\sqrt{1+x^2}} - \frac{x^2}{(1+x^2)}\frac{1}{2} + ...\right\} - \left\{x - \frac{x^2}{2} + ...\right\}}{\left[\frac{1}{2}\left\{x^2 - \frac{x^4}{2} + ...\right\} + \left\{\frac{x}{\sqrt{1+x^2}} - \frac{x^2}{(1+x^2)}\frac{1}{2} + ...\right\}\right]x}$$

$$= \lim_{x\to 0}\frac{\left(\frac{x}{\sqrt{1+x^2}} - x\right) + x^2 - x^2 - \frac{x^2}{2(1+x^2)} + ...}{\frac{x^2}{\sqrt{1+x^2}} + \frac{x^3}{2} - \frac{x^3}{(1+x^2)^2} + ...}$$

$$= \lim_{x\to 0}\frac{\left(1-\sqrt{1+x^2}\right)}{\sqrt{1+x^2}} + x - \frac{x}{2(1+x^2)} + ...$$

$$= \lim_{x\to 0}\frac{\frac{\left(1-\sqrt{1+x^2}\right)}{\sqrt{1+x^2}} + x - \frac{x}{2(1+x^2)} + ...}{\frac{x}{\sqrt{1+x^2}} + \frac{x^3}{2} - \frac{x^2}{(1+x^2)2} + ...}$$

$$= \lim_{x\to 0} \frac{\frac{-k}{\sqrt{1+x^2}\left[1+\sqrt{1+x^2}\right]} + 1 - \frac{1}{2\left(1+x^2\right)} + \text{term of } x}{\frac{1}{\sqrt{1+x^2}} + \text{term of } x}$$

$$= \frac{0+1-\frac{1}{2}+0}{1+0} = \frac{1}{2}$$

$$\frac{L+153}{L} = 1+306 = 307$$

21. Let $f(x) = \underset{n\to\infty}{\text{Lim}} \frac{2x^{2n}\sin\frac{1}{x}+x}{1+x^{2n}}$ then find.

यदि $f(x) = \underset{n\to\infty}{\text{Lim}} \frac{2x^{2n}\sin\frac{1}{x}+x}{1+x^{2n}}$ तो निम्नलिखित का मान ज्ञात कीजिए।

(a) $\underset{x\to\infty}{\text{Lim}}\, x f(x)$, **(b)** $\underset{x\to 1}{\text{Lim}}\, f(x)$, **(c)** $\underset{x\to 0}{\text{Lim}}\, f(x)$, **(d)** $\underset{x\to-\infty}{\text{Lim}}\, f(x)$

Ans. (a) 2, (b) D.N.E., (c) 0, (d) 0

Sol. $$f(x) = \begin{cases} x & x\in(-1,1) \\ 2\sin\left(\frac{1}{x}\right) & x\in(-\infty,-1)\cup(1,\infty) \\ \frac{2\sin 1+1}{2} & x=1 \\ \frac{-2\sin 1-1}{2} & x=-1 \end{cases}$$

(a) $\lim_{x\to\infty} x.f(x) - \lim_{x\to\infty} 2.\frac{\sin\left(\frac{1}{x}\right)}{\frac{1}{x}} = 2$

(b) $\lim_{x\to 1^-} f(x) = 1 = \text{L.H.L.} \quad \lim_{x\to 1} f(x) = \text{D.N.E.}$

$\lim_{x\to 1^+} f(x) = 2\sin 1 = \text{R.H.L.}$

(c) $\lim_{x\to 0} f(x) = \lim_{x\to 0} x = 0$

(d) $\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} 2\sin\left(\frac{1}{x}\right) = 0$

22. If $\lim_{x\to0}\frac{f(x)}{\sin^2 x} = 8$, $\lim_{x\to0}\frac{g(x)}{2\cos x - xe^x + x^3 + x - 2} = \lambda$ and $\lim_{x\to0}\left(1+2f(x)\right)^{\frac{1}{g(x)}} = \frac{1}{e}$, then $\lambda =$

यदि $\lim_{x\to0}\frac{f(x)}{\sin^2 x} = 8$, $\lim_{x\to0}\frac{g(x)}{2\cos x - xe^x + x^3 + x - 2} = \lambda$ तथा $\lim_{x\to0}\left(1+2f(x)\right)^{\frac{1}{g(x)}} = \frac{1}{e}$, है तो $\lambda =$

Ans. 8

Sol. $\lim_{x\to0}\frac{f(x)}{\sin^2 x} = 8$ if $\lim_{x\to0} f(x) \neq 0$

then L.H.S. $= \to\infty$

it means $\Rightarrow \lim_{x\to0} f(x) = 0$

$$\lim_{x\to0}\frac{f(x)}{\sin^2 x} = \lim_{x\to0}\frac{f(x)}{x^2} = 8$$

$$\lim_{x\to0}\frac{g(x)}{2\cos x - x.e^x + x^3 + x - 2} = \lambda$$

it is possible when $\lim_{x\to0} g(x) = 0$

Now

$$\lim_{x\to0}\left[1+2f(x)\right]^{\frac{1}{g(x)}} = \frac{1}{e} \qquad (1^\infty)$$

$$\Rightarrow e^{\lim_{x\to0}\frac{2f(x)}{g(x)}} = e^{-1}$$

$$\Rightarrow \lim_{x\to0}\ 2.\frac{f(x)}{g(x)} = -1$$

$$\Rightarrow \lim_{x\to0}\ 2.\frac{f(x)}{x^2}\frac{x^2}{g(x)} = -1$$

$$\lim_{x\to0}\ 2.8.\frac{x^2}{2\cos x - x.e^x + x^3 + x - 2}\frac{2\cos x - x.e^x + x^3 + x - 2}{g(x)} = -1$$

$$\Rightarrow \lim_{x\to0} 16.\frac{1}{\frac{-2(1-\cos x)}{x^2} - \frac{x\left(e^x-1\right)}{x^2} + x}\frac{1}{\lambda} = -1$$

$$\Rightarrow \frac{16}{-2\left(\frac{1}{2}\right)-1+0} = -\lambda \quad \Rightarrow \lambda = 8$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 23 to 25

Given $g(x) = \sin\frac{1}{x}$, $h(x) = x^2 + 2x + (\lambda + 1)$ and $u(x) = \frac{1}{x} + \cos\frac{1}{x^2}$

Let $f(x) = \lim_{n\to\infty} \frac{x^{2n+1}g(x) + h(x)}{x^{2n} + 3x.u(x)}$

On the basis of above information, answer the following questions :

दिया गया है $g(x) = \sin\frac{1}{x}$, $h(x) = x^2 + 2x + (\lambda + 1)$ तथा $u(x) = \frac{1}{x} + \cos\frac{1}{x^2}$

माना $f(x) = \lim_{n\to\infty} \frac{x^{2n+1}g(x) + h(x)}{x^{2n} + 3x.u(x)}$

उपरोक्त जानकारी के आधार पर निम्न प्रश्नों के उत्तर दीजिए :

23. If $\lim_{x\to 0} f(x) = 2$, then the value of λ is :

(A) 10 (B) 5 (C) 2 (D) Does not exist

यदि $\lim_{x\to 0} f(x) = 2$, तो λ होगा :

(A) 10 (B) 5 (C) 2 (D) विद्यमान नहीं है

Ans. B

Sol. $$f(x) = \lim_{n\to\infty} \frac{x^{2n+1}g(x) + h(x)}{x^{2n} + 3\times 4(x)} = \begin{cases} x.\sin\left(\frac{1}{x}\right) & x \in (-\infty,-1)\cup(1,\infty) \\ \frac{(x+1)^2 + \lambda}{3 + 3x\cos\left(\frac{1}{x^2}\right)} & x \in (-1,1) \\ \frac{\sin 1 + 4 + \lambda}{4 + 3\cos 1} & x = 1 \\ \frac{4 + \lambda - \sin 1}{4 - 3\cos 1} & x = -1 \end{cases}$$

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{(x+1)^2+\lambda}{3+3x\cos\left(\frac{1}{x^2}\right)} = 2$$

$$\Rightarrow \frac{1+\lambda}{3} = 2$$

$\lambda = 5$

24. $\lim_{x\to\infty} f(x) =$

(A) 0 (B) 2 (C) 5 (D) 1

Ans. D

Sol. $\lim_{x\to\infty} x\cdot f(x) = \lim_{x\to\infty} x\cdot\sin\left(\frac{1}{x}\right) = 1$

25. If $\lim_{x\to -2} f(x) = L$, then [L] (where [.] denotes greatest integer function) is equal to :

यदि $\lim_{x\to -2} f(x) = L$, तो [L] होगा : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

(A) 0 (B) 1 (C) –1 (D) 2

Ans. A

Sol. $L = \lim_{x\to -2} f(x) = \lim_{x\to -2} x.\sin\left(\frac{1}{x}\right) = 2.\sin\left(\frac{1}{2}\right) < 2.\left(\frac{1}{2}\right) < 2.\left(\frac{1}{2}\right) < 1$

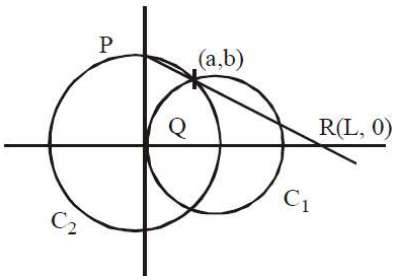
$[L] = 0$

Paragraph for Question Nos. 26 to 28

The figure shows a fixed circle C_1 and shrinking circle C_2. The equation of the fixed circle is $(x-1)^2 + y^2 = 1$. The centre of shrinking circle C_2 is at origin and it cuts positive y-axis at P. The points of intersection of C_1 and C_2 is Q (a, b) where $b > 0$, PQ cuts x-axis at R(L, 0). Let r be the radius of circle C_2 and let $L = f(r)$. On the basis of above information, answer the following questions :

चित्र में एक अचर वृत C_1 तथा चर वृत C_2 दिया गया है। वृत C_1 का समीकरण $(x-1)^2 + y^2 = 1$ है तथा वृत C_2 का केन्द्र मूल बिन्दु है तथा यह धनात्मक y-अक्ष को P पर काटता है। C_1 तथा C_2 का प्रतिछेदन बिन्दु Q (a, b) है जहाँ $b > 0$, PQ x-अक्ष को R(L, 0) पर काटता है। माना वृत C_2 की त्रिज्या r है तथा $L = f(r)$ ।

उपरोक्त जानकारी के आधार पर निम्न प्रश्नों के उतर दीजिए।

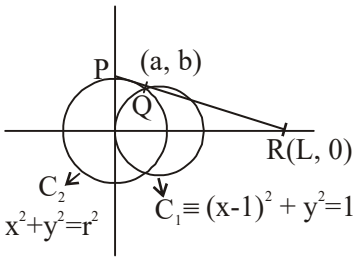


26. $\lim_{r\to 0} f(r) =$

(A) 0 (B) 2 (C) 4 (D) 8

Ans. C

Sol.



P(0, r)

$a^2 + b^2 = r^2$ _____(1)

$(a-1)^2 + b^2 = 1$ _____(2)

(1) and (2) $2a - 1 = r^2 - 1$ $\Rightarrow a = \frac{r^2}{2}$

$b^2 = r^2 - \frac{r^4}{4} = r^2\left(4 - r^2\right)$

$b = \frac{r\sqrt{4-r^2}}{2}$ equation of PR $\frac{x}{L} + \frac{y}{r} = 1$

(a, b) lies on PR $\frac{a}{L} + \frac{b}{r} = 1$

$\Rightarrow \frac{r^2}{2L} + \frac{\sqrt{4-r^2}}{2} = 1$

$\Rightarrow \frac{L}{r^2} = \frac{1}{2-\sqrt{4-r^2}} = \frac{2+\sqrt{4-r^2}}{r^2}$

$\Rightarrow L = 2 + \sqrt{4-r^2} = f(r)$

Now $\lim_{r\to 0} f(r) = 2 + \sqrt{4-0} = 4$

27. If $f(r).g(r) = r^2$, then $\lim_{\theta\to 0}\frac{g(2\sin\theta)}{\theta^2}$ is :

यदि $f(r).g(r) = r^2$, तो $\lim_{\theta\to 0}\frac{g(2\sin\theta)}{\theta^2}$ होगा :

(A) 0 (B) 1 (C) 1/2 (D) 2

Ans. B

Sol. $f(r).g(r) = r^2 \Rightarrow g(r) = \frac{r^2}{2+\sqrt{4-r^2}}$

$$g(2\sin\theta) = \frac{4\sin^2\theta}{2+2\cos\theta}$$

$$\frac{g(2\sin\theta)}{\theta^2} = \frac{2\sin^2\theta}{\theta^2}\cdot\frac{1}{1+\cos\theta}$$

$$\lim_{\theta\to 0}\frac{g(2\sin\theta)}{\theta^2} = 2\frac{1}{1+1} = 1$$

28. Range of f is :

f का परिसर होगा :

(A) [2, 4] (B) (2, 4] (C) [2, 4) (D) (2, 4)

Ans. D

Sol. $f(r) = 2+\sqrt{4-r^2}$

$\because -r^2 < 0 \quad \Rightarrow 0 < \sqrt{4-r^2} < 2$

$2 < 2+\sqrt{4-r^2} < 4$

Range $\in (2, 4)$

Paragraph for Question Nos. 29 to 31

Let f(x) is a function continuous for all $x \in R$ except at $x = 0$. Such that $f'(x) < 0 \ \forall\ x \in (-\infty, 0)$ and $f'(x) > 0$ $\forall\ x \in (0, \infty)$. Let $\lim\limits_{x\to 0^+} f(x) = 2$, $\lim\limits_{x\to 0^-} f(x) = 3$ and $f(0) = 4$.

माना फलन f(x), $x = 0$ को छोड़कर सभी $x \in R$ के लिए इस प्रकार सतत है, कि $f'(x) < 0 \ \forall\ x \in (-\infty, 0)$ तथा $f'(x) > 0 \ \forall\ x \in (0, \infty)$. माना $\lim\limits_{x\to 0^+} f(x) = 2$, $\lim\limits_{x\to 0^-} f(x) = 3$ तथा $f(0) = 4$.

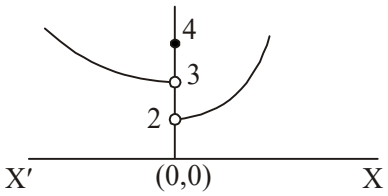
29. The value of λ for which $2\left(\lim_{x\to 0} f(x^3 - x^2)\right) = \lambda\left(\lim_{x\to 0} f(2x^4 - x^5)\right)$ is :

λ का मान होगा जिसके लिए $2\left(\lim\limits_{x\to 0} f(x^3-x^2)\right)=\lambda\left(\lim\limits_{x\to 0} f(2x^4-x^5)\right)$:

(A) $\frac{4}{3}$ (B) 2 (C) 3 (D) 5

Ans. C

Sol.



$x \to 0^-$ $x^3 = -ve$ $x^2 = +ve$ $x^3 - x^2 < 0$

$\Rightarrow f(X^3 - X^2) > f(0)$

$\lim\limits_{x\to 0^-} f\left(X^3-X^2\right)=3^+$ $f(x^3 - x^2) > 4$

$=\lim\limits_{x\to 0^-}\left[f\left(X^3-X^2\right)\right]=3$

$x \to 0^+$ $2x^4 - x^5 > 0$ $x \to 0^+$ $f(2x^4 - x^5) = 2^+$

$f(2x^4 - x^5) > f(0)$

$f(2x^4 - x^5) > 4$ $=\lim\limits_{x\to 0^+}\left[f\left(2x^4-x^5\right)\right]=2$

$$2\left[\lim_{x\to 0} f\left(x^3-x^2\right)\right]=\lambda\left[\lim_{x\to 0} f\left(2x^4-x^5\right)\right]$$

$$\Rightarrow 2\left[\lim_{x\to 0^-} f\left(x^3-x^2\right)\right]=\lambda\left[\lim_{x\to 0^+} f\left(2x^4-x^5\right)\right]$$

$\Rightarrow 2(3) = \lambda(2) \Rightarrow \lambda = 3$

30. The values of $\lim\limits_{x\to 0^+} \dfrac{f(-x)\,x^2}{\left\{\dfrac{1-\cos x}{[f(x)]}\right\}}$ where [·] denote greatest integer function and { · } denote fraction part function:

$\lim\limits_{x\to 0^+} \dfrac{f(-x)\,x^2}{\left\{\dfrac{1-\cos x}{[f(x)]}\right\}}$ का मान होगा जहां [·] महत्तम पूर्णांक फलन तथा { · } भिन्नात्मक भाग फलन को दर्शाता है :

(A) 6 (B) 12 (C) 18 (D) 24

Ans. B

Sol. $x \to 0^+ \Rightarrow -x \to 0^-$ $\lim\limits_{x\to 0^+} f(-x) = 3$

$\because \lim\limits_{x\to 0^-} f(x) = 3$ $[f(x)] = 2$

$x \to 0^+$ $0 < \dfrac{1-\cos x}{2} < 1 \Rightarrow \left\{\dfrac{1-\cos x}{2}\right\} = \dfrac{1-\cos x}{2}$

$$\lim_{x\to 0^+} \frac{f(-x)x^2}{\left\{\frac{1-\cos x}{2}\right\}} = 3\cdot 2\cdot \lim_{x\to 0^+} \frac{x^2}{1-\cos x} = 12$$

31. $\lim\limits_{x\to 0^-} \left(\left[3f\left(\dfrac{x^3-\sin^3 x}{x^4}\right)\right] - f\left(\left[\dfrac{\sin x^3}{x}\right]\right)\right)$ where $[\cdot]$ denote greatest integer function :

$\lim\limits_{x\to 0^-} \left(\left[3f\left(\dfrac{x^3-\sin^3 x}{x^4}\right)\right] - f\left(\left[\dfrac{\sin x^3}{x}\right]\right)\right)$ है जहां $[\cdot]$ महत्तम पूर्णांक फलन को दर्शाता है :

(A) 3 (B) 5 (C) 7 (D) 9

Ans. B

Sol. $x \to 0^-$ $x^3 < \sin^3 x$ $\dfrac{x^3-\sin^3 x}{x^4} < 0$

$$\Rightarrow f\left(\frac{x^3-\sin^3 x}{x^4}\right) = 3^+$$

$$\Rightarrow \left[f\left(\frac{x^3-\sin^3 x}{x^4}\right)\right] = 3$$

and

$x \to 0^-$ $\sin^3 x > x$ $\left[\dfrac{\sin^3 x}{x}\right] = 0$ $f\left(\left[\dfrac{\sin^3 x}{x}\right]\right) = f(0) = 4$

Now

$$\lim_{x\to 0^-} \left[3f\left(\frac{x^3-\sin^3 x}{x^4}\right)\right] - f\left(\left[\frac{\sin^3 x}{x}\right]\right)$$

$= 9 - 4 = 5$

MATCH THE COLUMN

32. Match the following column :

Column-I	**Column-II**
(A) $\lim_{n\to\infty}\left[\sum_{r=1}^{n}\frac{1}{2^r}\right]$ is equal to	(P) 0
(B) $\lim_{x\to\frac{\pi}{2}}(\sin x)^{\frac{1}{2x-\pi}}$	(Q) 1
(C) $\lim_{n\to\infty}\left[n^2\sin\left(\pi\cos^2\left(\tan\sin\frac{1}{n}\right)\right)\right]$ is equal to	(R) 2
(D) The absolute value of $\lim_{x\to 0}\frac{\sqrt{1+\sin^2 x}-\cos x}{1-\sqrt{1+\tan^2 x}}$ is	(S) 3

(where [.] denotes greatest integer function)

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Column-I	**Column-II**
(A) $\lim_{n\to\infty}\left[\sum_{r=1}^{n}\frac{1}{2^r}\right]$	(P) 0
(B) $\lim_{x\to\frac{\pi}{2}}(\sin x)^{\frac{1}{2x-\pi}}$	(Q) 1
(C) $\lim_{n\to\infty}\left[n^2\sin\left(\pi\cos^2\left(\tan\sin\frac{1}{n}\right)\right)\right]$	(R) 2
(D) $\lim_{x\to 0}\frac{\sqrt{1+\sin^2 x}-\cos x}{1-\sqrt{1+\tan^2 x}}$ का निरपेक्ष मान होगा	(S) 3

(जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

Ans. A → P, B → Q, C → S, D → R

Sol. (A) $\lim_{n\to\infty}\left[\sum_{r=1}^{n}\frac{1}{2^r}\right]=\lim_{n\to\infty}\left[\frac{1}{2}\left(\frac{1-\left(\frac{1}{2}\right)^n}{1-\frac{1}{2}}\right)\right]$

$=\lim_{n\to\infty}\left[1-\left(\frac{1}{2}\right)^n\right]=\lim_{n\to\infty}0$

Hence A → P

(B) $\lim\limits_{x\to\frac{\pi}{2}}(\sin x)^{\frac{1}{2x-\pi}}$ $[1^\infty]$

$= e^{\lim\limits_{x\to\frac{\pi}{2}}\frac{\sin x-1}{2x-\pi}} = e^0 = 1$

$B \to Q$

(C) $\lim\limits_{n\to\infty}\left[n^2\sin\left(\pi-\pi\cos^2\left(\tan\sin\frac{1}{n}\right)\right)\right]$

$= \lim\limits_{n\to\infty}\left[n^2\sin\left(\pi\sin^2\left(\tan\left(\sin\left(\frac{1}{n}\right)\right)\right)\right)\right]$

$= \lim\limits_{n\to\infty}\left[n^2\dfrac{\pi\cdot\sin\left(\pi\sin^2\left(\tan\left(\sin\left(\frac{1}{n}\right)\right)\right)\right)}{\pi\sin^2\left(\tan\left(\sin\frac{1}{n}\right)\right)}\cdot\dfrac{\sin^2\tan\left(\sin\left(\frac{1}{n}\right)\right)}{\tan^2\sin\left(\frac{1}{n}\right)}\right]$

$= [\pi] = 3$ $\qquad C \to S$

(D) $\lim\limits_{x\to0}\dfrac{\sqrt{1+\sin^2 x}-\cos x}{1-\sqrt{1+\tan^2 x}}$

$= \lim\limits_{x\to0}\dfrac{2\sin^2 x}{-\tan^2 x}\cdot\dfrac{1+\sqrt{1+\tan^2 x}}{\sqrt{1+\sin^2 x}+\cos x}$

$= -2\cdot\dfrac{2}{2}$

$\Rightarrow$ absolute value = 2

33. Match the following column :

Column-I	**Column-II**
(A) $\lim\limits_{n\to\infty}\cos^2\left(\pi\left(\sqrt[3]{n^3+n^2+2n}\right)\right)$, where *n* is an integer, equals	(P) $\frac{1}{2}$
(B) $\lim\limits_{n\to\infty}\ n\sin\left(2\pi\sqrt{1+n^2}\right)$ $(n\in N)$ equals	(Q) $\frac{1}{4}$
(C) $\lim\limits_{n\to\infty}(-1)^n\sin\left(\pi\sqrt{n^2+0.5n+1}\right)\left(\sin\frac{(n+1)\pi}{4n}\right)$ is (where $n\in N$)	(R) π

(D) If $\lim\limits_{x\to\infty}\left(\frac{x+a}{x-a}\right)^x = e$ where 'a' is some real constant then the value of 'a' is equal to — (S) non existent

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) $\lim\limits_{n\to\infty}\cos^2\left(\pi\left(\sqrt[3]{n^3+n^2+2n}\right)\right)$, जहाँ n पूर्णांक है, का मान होगा	(P) $\frac{1}{2}$
(B) $\lim\limits_{n\to\infty} n\sin\left(2\pi\sqrt{1+n^2}\right)$ $(n \in N)$ का मान होगा	(Q) $\frac{1}{4}$
(C) $\lim\limits_{n\to\infty}(-1)^n\sin\left(\pi\sqrt{n^2+0.5n+1}\right)\left(\sin\frac{(n+1)\pi}{4n}\right)$ (जहाँ $n \in N$) का मान होगा	(R) π
(D) यदि $\lim\limits_{x\to\infty}\left(\frac{x+a}{x-a}\right)^x = e$ जहाँ 'a' कोई वास्तविक नियतांक है तब 'a' का मान होगा।	(S) अस्तित्व विहीन

Ans. $A \to Q; B \to R; C \to P; D \to P$

Sol. $\because \pi\left(n^3+n^2+2n\right)^{\frac{1}{3}}$

$$n\pi\left[1+\frac{n(n+2)}{n^3}\right]^{\frac{1}{3}}$$

$$= n\pi\left[1+\frac{n+2}{n^2}\cdot\frac{1}{3}+\frac{1}{3}\left(-\frac{2}{3}\right)\cdot\frac{1}{3!}\left(\frac{n+2}{n^2}\right)^2+.....\right]$$

$$= n\pi+\frac{n+2}{n}\cdot\frac{\pi}{3}-\underbrace{\frac{\pi}{9}\frac{(n+2)^2}{n^3}+.....}_{\text{in these terms power of } N^r < D^r}$$

Now

$$= \lim_{n\to\infty}\left[\cos\left(\pi\left(n^3+n^2+2n\right)^{\frac{1}{3}}\right)\right]^2$$

$$= \lim_{n\to\infty}\left[\cos\left[n\pi+\frac{n+2}{n}\frac{\pi}{3}-\frac{\pi}{9}\frac{(n+2)^2}{n^3}+.....\right]\right]^2$$

$$= \lim_{n\to\infty}\left[\pm\cos\left[\frac{n+2}{n}\frac{\pi}{3}-\frac{\pi}{9}\frac{(n+2)^2}{n^3}+.....\right]\right]^2$$

$$= \lim_{n\to\infty}\left[\pm\cos\left[\frac{n+2}{n}\frac{\pi}{3}-\frac{\pi}{9}\frac{(n+2)^2}{n^3}+....\right]\right]^2$$

$$=\left(\pm\cos\frac{\pi}{3}\right)^2$$

$$=\left(\pm\frac{1}{2}\right)^2=\frac{1}{4}$$

$A \to Q$

(B) $2\pi\left(1+n^2\right)^{\frac{1}{2}}$

$$= 2\pi n\left[1+\frac{1}{n^2}\right]^{\frac{1}{2}}$$

$$= 2n\pi\left[1+\frac{1}{2}\frac{1}{n^2}+\frac{1}{2}\left(\frac{1}{2}-1\right)\frac{1}{2!}\frac{1}{n^4}+.....\right]$$

$$=\left[2n\pi+\frac{\pi}{n}-\frac{\pi}{4}.\frac{1}{n^3}+....\right]$$

Now

$$\lim_{n\to\infty} n\sin\left(2\pi\sqrt{1+n^2}\right)$$

$$= \lim_{n\to\infty}\frac{\sin\left[2n\pi+\frac{\pi}{n}-\frac{\pi}{4}.\frac{1}{n^3}+....\right]}{\frac{1}{n}}$$

$$= \lim_{n\to\infty}\frac{\sin\left\{\frac{\pi}{n}-\frac{\pi}{4}.\frac{1}{n^3}+.....\right\}}{\left\{\frac{\pi}{n}-\frac{\pi}{4}.\frac{1}{n^3}+.....\right\}}.n\left\{\frac{\pi}{n}-\frac{\pi}{4}.\frac{1}{n^3}+....\right\}$$

$= 1 \cdot (\pi - 0) = \pi$

$B \to R$

(C) $\lim_{n\to\infty}(-1)^n \sin\left[n\pi\left(1+\frac{1+\frac{1}{2}n}{n^2}\right)^{\frac{1}{2}}\right]\sin\left(\frac{\pi}{4}+\frac{\pi}{4n}\right)$

$= \lim_{n\to\infty}(-1)^n \sin\left[n\pi\left(1+\frac{1}{2}\frac{1+\frac{1}{2}n}{n^2}+\left(-\frac{1}{4}\right)\cdot\frac{\left(1+\frac{1}{2}n\right)^2}{n^4}+\ldots\right)\right]\sin\frac{\pi}{4}$

$= \lim_{n\to\infty}(-1)^n\cdot(-1)^n \sin\left[\frac{\pi}{2}\frac{1+\frac{1}{2}n}{n}+\text{ term of }\frac{1}{n}\right]\cdot\sin\frac{\pi}{4}$

$= \sin^2\frac{\pi}{4} = \frac{1}{2}$

$C \to P$

(D) $\lim_{x\to\infty}\left(\frac{x+a}{x-a}\right)^x = e$

$= e^{\lim_{x\to\infty}\left(\frac{x+a}{x-a}-1\right)x} = e$

$= \lim_{x\to\infty}\frac{2ax}{x-a} = 1 \Rightarrow 2a = 1 \Rightarrow a = \frac{1}{2}$

$D \to P$

34. Match the following column :

Column-I	**Column-II**
(A) $\lim_{n\to\infty}\left(\frac{1+\sqrt[n]{4}}{2}\right)^n =$	(P) 2
(B) Let $f(x) = \lim_{n\to\infty}\frac{2x}{\pi}\tan^{-1}(nx)$, then $\lim_{x\to0^+} f(x) =$	(Q) 0
(C) $\lim_{x\to\frac{\pi^+}{2}}\frac{\cos\left(\tan^{-1}(\tan x)\right)}{x-\frac{\pi}{2}}$	(R) 1
(D) If $\lim_{x\to0^+}(x)^{\frac{1}{\ln\sin x}} = e^L$, then $L+2 =$	(S) 3
	(T) Non existent

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) $\lim\limits_{n\to\infty}\left(\dfrac{1+\sqrt[n]{4}}{2}\right)^n =$	(P) 2
(B) माना $f(x)=\lim\limits_{n\to\infty}\dfrac{2x}{\pi}\tan^{-1}(nx)$ है, तो $\lim\limits_{x\to0^+} f(x) =$	(Q) 0
(C) $\lim\limits_{x\to\frac{\pi^+}{2}}\dfrac{\cos\left(\tan^{-1}(\tan x)\right)}{x-\frac{\pi}{2}}$	(R) 1
(D) यदि $\lim\limits_{x\to0^+}(x)^{\frac{1}{\ln\sin x}}=e^L$ है, तो $L+2=$	(S) 3
	(T) विद्यमान नहीं

Ans. $A\to P;\ B\to Q;\ C\to R;\ D\to S$

Sol. (A) $\lim\limits_{n\to\infty}\left(\dfrac{1+4^{\frac{1}{n}}}{2}\right)^n$ 1^∞

$=e^{\lim\limits_{n\to\infty}\left(\frac{4^{\frac{1}{n}}-1}{2}\right)n}=e^{\lim\limits_{x\to0}\frac{4^x-1}{x}\cdot\frac{1}{2}}$ where $x=\dfrac{1}{n}$

$=e^{\ell n4\cdot\frac{1}{2}}=2$

$A\to P$

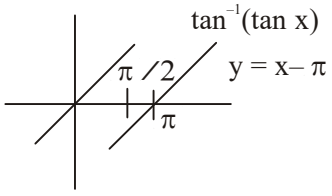
(B) $f(x)=\lim\limits_{n\to\infty}\dfrac{2x}{\pi}\tan^{-1}(nx)$ $\because x\to0^+$

$=\dfrac{2x}{\pi}\dfrac{\pi}{2}=x$

$f(x)=x$ $B\to Q$

$x\to0^+$ $f(x)\to0$

(C) $\lim\limits_{x\to\frac{\pi^+}{2}}\dfrac{\cos\left(\tan^{-1}(\tan x)\right)}{x-\frac{\pi}{2}}$



$$= \lim_{x\to\frac{\pi}{2}^+} \frac{\cos(k-\pi)}{x-\frac{\pi}{2}} = \lim_{x\to\frac{\pi}{2}^+} \frac{-\cos x}{-\left(\frac{\pi}{2}-x\right)}$$

$= 1$ $\qquad\qquad C \to R$

(D) $e^L = \lim_{x\to 0^+} (x)^{\frac{1}{n\sin x}}$

$$L = \lim_{x\to 0^+} \frac{1}{\ell n \sin x}\cdot \ell n x$$

By L-Hospital Rule

$$L = \lim_{x\to 0^+} \frac{\sin x}{x}\cdot \sec x = 1$$

$L + 2 = 3$